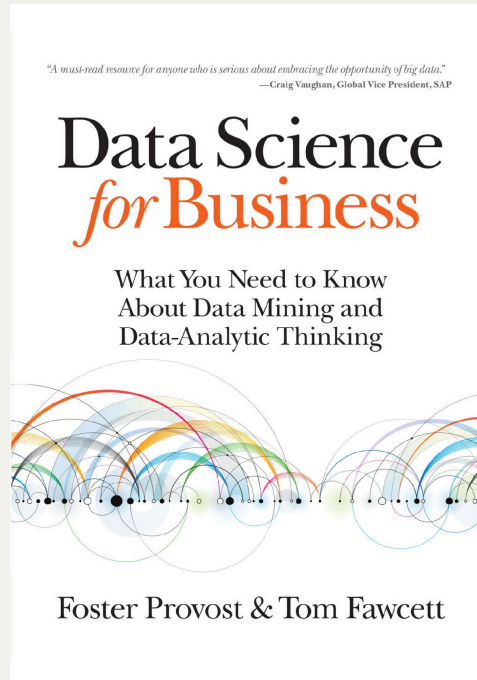


Data Science as 9 problems

AP

A gentle-yet-focussed introduction

Chapter 2 describes the core computational problems of Data Science



In memoriam: Tom Fawcett

The big 9

1. Classification and class probability

Instance:

- a collection (dataset) $\mathbf{D}$ of datapoints from $\mathbf{X}$, e.g., $\mathbb{R}^m$
- a classification system $C = \{c_1, c_2, \dots, c_k\}$

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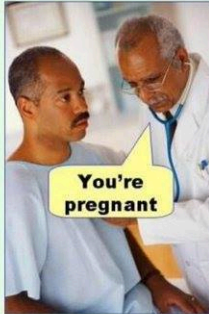
Solution: classification function $\gamma : \mathbf{X} \rightarrow C$

Measure: misclassification of datapoints in **D**

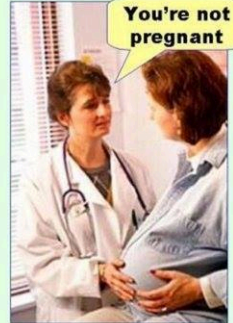
Misclassification: detection

binary, only one class is important

Type I error
(false positive)



Type II error
(false negative)



Misclassification: multi-class

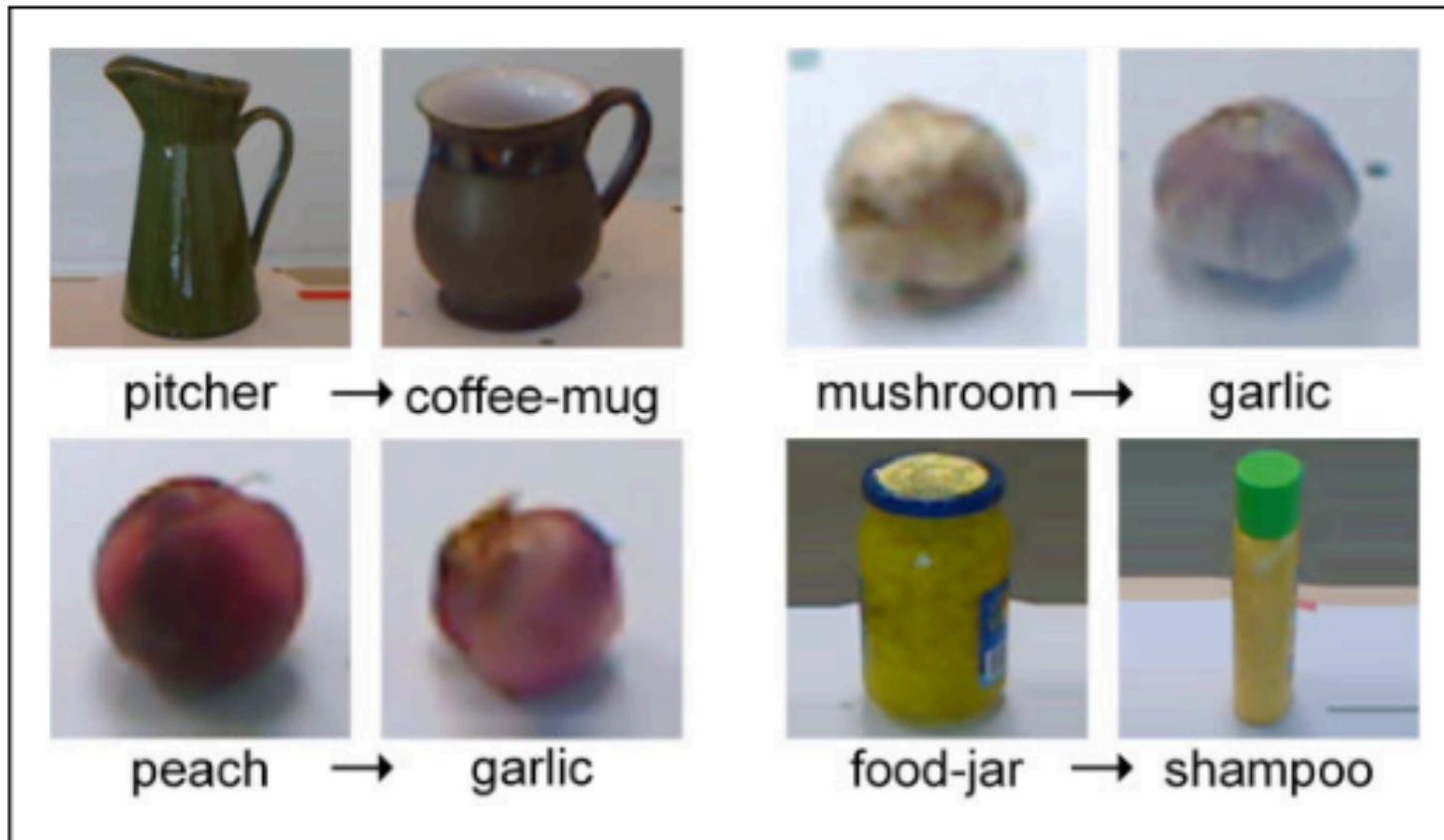
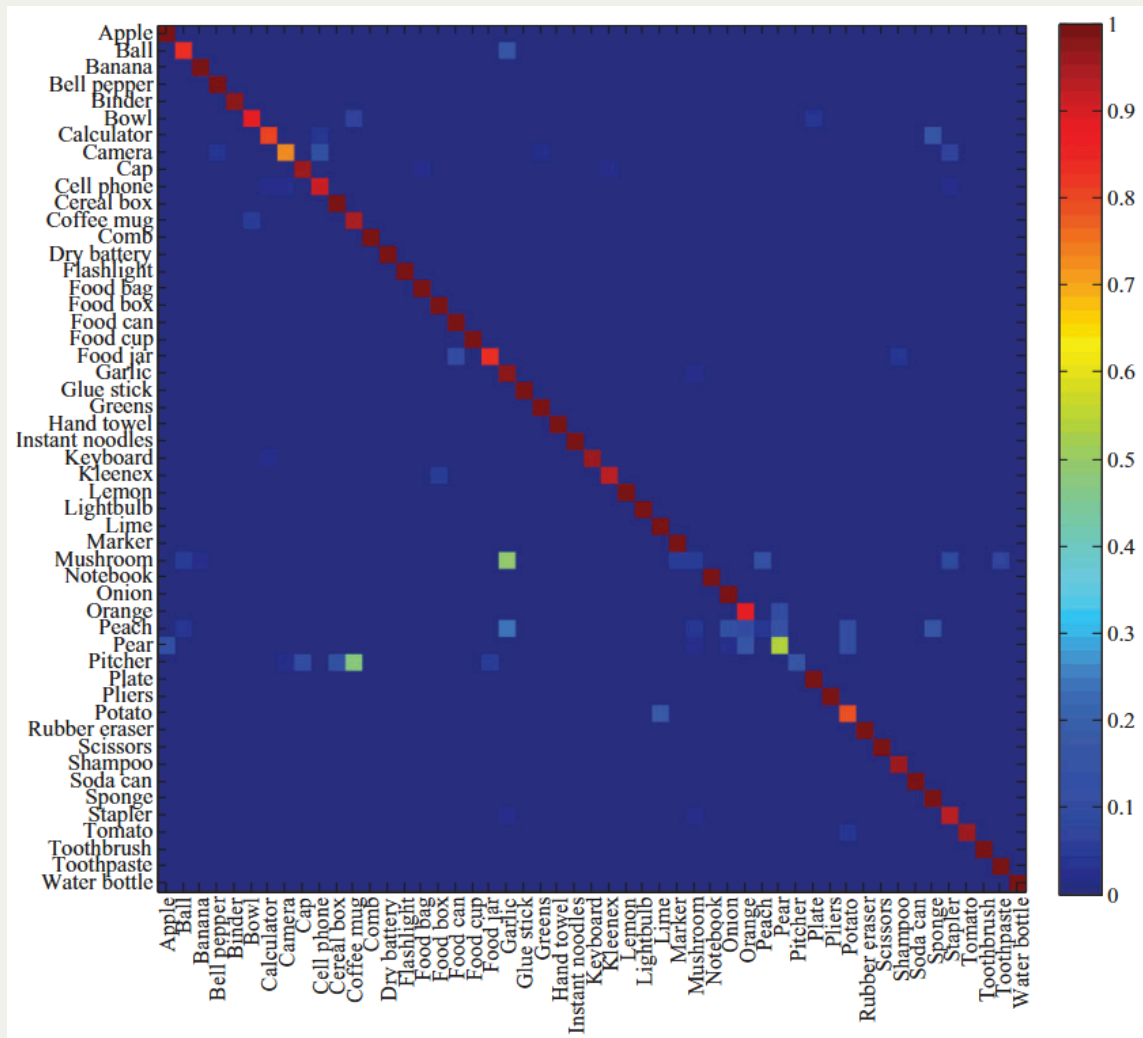


Figure 8. Examples of some easily misclassified categories. Misclassification occurs due to the strong similarities in the objects' colours and shapes.



Source

2. Regression/value estimation

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[PF] “classification predicts whether something will happen, whereas regression predicts how much something will happen.”

3. Similarity

Identify similar individuals based on data known about them.

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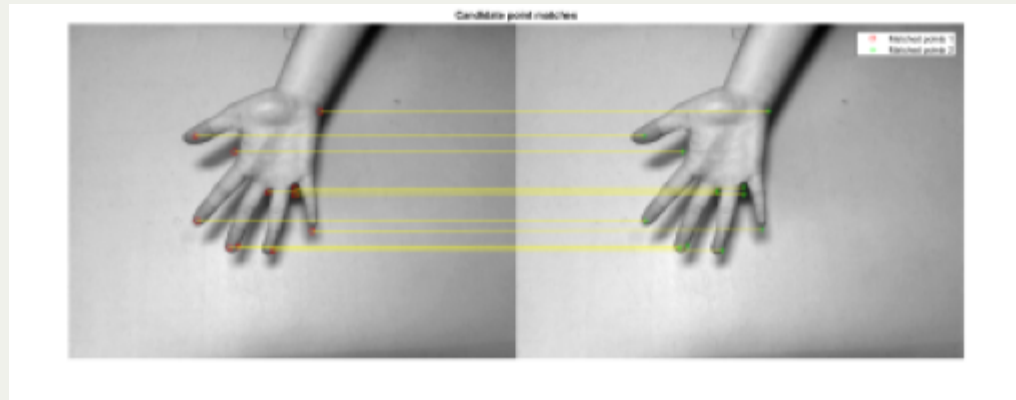
Instance:

- a collection (dataset) $\mathbf{D}$ of datapoints from $\mathbf{X}$, e.g., $\mathbb{R}^m$
- (distance functions for some of the dimensions)

Solution: similarity function $\sigma : \mathbf{X} \rightarrow \mathbb{R}$

[**Measure:** error]

Good similarity measures are the key to accurate detection/classification



4. Clustering (segmentation)

group individuals in a population together by their similarity (but not driven by any specific purpose)

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- a collection (dataset) $\mathbf{D}$ of datapoints from $\mathbf{X}$, e.g., $\mathbb{R}^m$
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- a small integer k

Solution: a partition of $\mathbf{D}$ into $\mathcal{C}_1, \dots, \mathcal{C}_k$

Measure: network modularity Q : proportion of the relational structure that *respects* the clusters.

Detection version: k is part of the output.

See an example research work (from yours truly)

5. Co-occurrence (frequent itemset mining)

detect similarities of objects on the basis on their appearing together in transactions.

Instance:

- a collection (dataset) $\mathbf{T}$ of *itemsets* (subsets of $\mathbf{X}$) (or sequences)
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Market-basket analysis, some recommendation systems

6. Profiling (behavior description)

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- a stimulus $a \in \mathbf{A}$
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Application: anomaly/fraud detection.

Example: Social media profiling

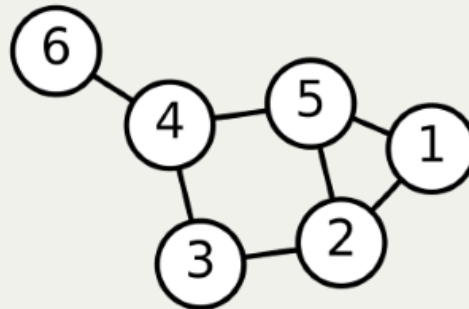
7. Link prediction

Instance: a dynamical graph (network) $\mathbf{G}$, i.e., a sequence

$$\langle V, E \rangle,$$

$$\langle V, E' = E + \{(u, v)\} \rangle,$$

$$\langle V, E'' = E' + \{(r, s)\} \rangle \dots$$



Question: what is the next link to be created?

What YouTube video will you watch next?

Alternatives: predict the **strength** of the new link; link deletion.

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- [a distinct independent variable x_i]

Solution: a projection of **D** onto $\mathbb{R}^n, n < m$

Measure: error in the estimation of x_i

Example: genre identification in consumer behaviour analysis

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Example: Exactly What food causes you to put on weight?

Controlled clinical trials, A/B testing.

From problems to algorithms

Computer Science is often metaphoric: it uses abstractions that allow us to

- focus our problem-solving
- communicate with the computer

Examples:

- interpreted languages (Python)
- data models (SQL and relational tables)
- TCP/IP computer networks

Problem $\longrightarrow$ Algorithm $\longrightarrow$ Implementation (code)

For a given problem, more than one algorithm may be available

For a given algorithm, more than one implementation is possible

Only with clarity about the problem we can look for the algorithms.

Supervised Algorithms

Previous instance/solution pairs are available and fed to the a.

The a. may 'memorise' past solutions and re-apply them, via some similarity criterion.

Advanced a. may 'learn' a model and apply it to future inputs.

Method

- obtain a dataset of *labeled* examples: the class assignment or the value are known;
- split it in training and test data;
- run a. on the test data, find a putative solution, and
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Regression has a numeric target while classification has a categorical/binary one

P. with good supervised algorithms

1: Regression

2: Classification

9: Causal Modelling

P. with mostly unsupervised a.

4: Clustering

5: co-occurrence grouping

6: profiling

P. with a mix of supervised and unsupervised

3: Similarity matching,

7: link prediction,

8: data reduction